

File Descriptor	Abrr	Symbol	Description
0	STDIN	<	Standard input (e.g what you type in the terminal. When you use < file Linux read the file and retrieves data in the same way that you would type it on a keyboard.
1	STDOUT	>	Standard Output e.g terminal screen by default. All programs and scripts output is redirected to the terminal screen. When you use > Linux replaces the terminal with a file referenced by the redirection symbol.
2	STDERR	2>	Standard Error. By default, points to the same location as the STDOUT file descriptor (despite being assigned different file descriptor values). This means that by default, all error messages are displayed on the monitor.

0

/dev/tty0

1

/dev/tty0

2

/dev/tty0



Unless they are closed, file descriptors always point to a file. When bash starts, all three file descriptors, **STDIN** (file descriptor 0), **STDOUT** (file descriptor 1), and **STDERR** (file descriptor 2), usually point to your terminal. In this case our terminal is **/dev/tty0**

You can also open and close additional file descriptors (such as 3, 4, 5,...). File descriptors can also be copied. You can also write to and read from them.

Simple Redirections

Input Redirection

0

/opt/data/txt

1

/dev/tty0

2

/dev/tty0

Here bash changes the file descriptor 0 to point to a file **/opt/data/txt**

\$ command < /opt/data.txt

This is the same as <0

Output Redirection

0

/dev/tty0

1

/opt/data/txt

2

/dev/tty0

Here bash changes the file descriptor 1 to point to a file **/opt/data/txt**

\$ command > /opt/data.txt

This is the same as 1>

Error Redirection

0

/dev/tty0

1

/dev/tty0

2

/opt/data/txt

Here bash changes the file descriptor 2 to point to a file **/opt/data/txt**

\$ command 2> /opt/data.txt

 In general you can write command [n]> file, which will redirect the file descriptor n to file

Redirecting both STDOUT and STDERR to a File

\$ command > /dev/null 2>&1

0

/dev/tty0

1

/dev/tty0

2

/dev/tty0

redirection process

0

/dev/tty0

1

/dev/null

2

/dev/tty0

\$ command > /dev/null

Here bash changes the file descriptor 1 to point to a file **/dev/null**

0

/dev/tty0

1

/dev/null

2

/dev/null

\$ command > /dev/null 2>&1

Here bash duplicates file descriptor 2 to be a copy of file descriptor 1

duplication

 The above can be also written as **command &> /dev/null** or command **command >& /dev/null**. Note: Redirections are processed from left to right and order is very important

\$ command > /dev/null 2>&1 is not equal to \$ command 2>&1 > /dev/null

0

/dev/tty0

1

/dev/tty0

2

/dev/tty0

\$ command 2>&1

Here bash duplicates file descriptor 2 to be a copy of file descriptor 1. At this point 1 was still pointing to **/dev/tty0**

0

/dev/tty0

1

/dev/tty0

2

/dev/null

\$ command 2>&1 > /dev/null

Here bash changes the file descriptor 1 to point to a file **/dev/null**